CHU NGOC VUONG

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SUMMARY

AI engineer with 2 years building production LLM systems — retrieval-augmented search, multi-agent pipelines, and the serving infrastructure around them. Core engineer on the RAG platform that won a client engagement; currently own the retrieval and tariff-resolution layer of a US customs classification product, and have delivered onsite to a Danish client. Published at VLSP 2025 (ACL), placed first at ImageCLEFmed MEDIQA-CORE 2026, and research AI for health — dementia and Alzheimer's disease.

EDUCATION

University of Engineering and Technology (UET), Vietnam National University Hanoi, Vietnam
B.Sc. in Computer Science; GPA 3.74 / 4.00 September 2021 – 2025

EXPERIENCE

NTQ Solution JSC Hanoi, Vietnam
AI Engineer July 2024 – Present

US HTS Code Search & Classification Platform May 2026 – Present

- Reached **~99% top-3 recall** on a labeled set by adding an LLM pre-search selector that hard-filters retrieval to the top-3 tariff chapters — off the event loop, timeout-bounded and fail-open, so a slow LLM degrades rather than fails.
- Cut tariff lines unresolvable for a non-FTA origin **from 55.2% to 7.9%** by implementing country-of-origin duty resolution across GSP, AGOA, CBTPA, CAFTA-DR, CBERA and USMCA; found and fixed a high-severity bug that granted duty-free treatment under LDC-only symbols to non-LDC countries.
- Own the retrieval layer classifying free-text product descriptions into 10-digit US Harmonized Tariff Schedule codes: hybrid BM25 + kNN search over OpenSearch, merged by weighted Reciprocal Rank Fusion across a 23,392-document index.
- Made reindexing zero-downtime with blue/green index builds and atomic alias swaps; keep a 190+ test suite mypy- and ruff-clean in GitLab CI.

Multi-Agent RAG Portal for Proteomics Reports — Biotech Client March 2026 – June 2026

- Replaced a single-shot RAG call with an **8-agent orchestration pipeline** (planner, decomposition, HyDE, retrieval, synthesis, fact-check, memory) in which every stage is runtime-toggleable by feature flag, so agents could be compared and rolled back in production without a redeploy.
- Cut ingested report size **97% (830 KB → 20 KB)** by stripping base64 images before chunking, and made tables answerable by indexing each as cross-linked summary and raw-markdown chunks — a semantic hit on the summary pulls the exact original table rather than a split fragment.
- Owned the retrieval and ingestion backend as the AI lead on a 4-engineer team: OpenSearch kNN (HNSW/cosine, 1536-dim) with per-tenant filtering, asynchronous Celery ingestion, and Claude-backed synthesis streamed over SSE; shipped to production on AWS ECS Fargate.
- Kept evaluation honest with a 3-layer harness separating deterministic ingestion invariants from LLM-quality judging, so artifact diffs distinguish real regressions from model sampling noise.

CRM Documentation Platform — Danish Maritime Client Copenhagen, Denmark · August 2025 – May 2026

- Replaced a manual, hand-written BA documentation process with a 12-agent pipeline turning user stories or source code into structured business and API documentation — **30+ entity documents** plus cross-entity flow, integration and permission models.
- Kept generated docs reviewable rather than merely plausible: quality gates halt on ambiguous input and a bounded revision loop runs before escalating to a human.
- Made documentation regenerable from current implementation via a code-reader agent that indexes the .NET backend and React/Next.js frontend into a codemap.
- Delivered onsite in Copenhagen for 2 weeks and ran requirements and review conversations with the client in English over roughly 6 months.

NxKMS — AI Knowledge Management System July 2024 – July 2025

- Core engineer on the RAG platform that **won the client engagement** — the most complex system I have worked on: multi-agent, multi-step, message-queue driven.
- Kept large-document ingestion off the request path with asynchronous Celery and RabbitMQ processing for PDF, DOCX, XLSX, PPTX, TXT and JSON, behind a strategy pattern that admits new processors without touching orchestration.
- Built the custom hybrid search engine with reranking over Qdrant embeddings and a 4-language prompt system (EN/JA/KO/VI), plus the FastAPI service, its PostgreSQL/SQLAlchemy/Alembic and MinIO data layer, and the Docker packaging that made dev, staging and production reproducible.

Graph-Based Retrieval for Technical Documentation December 2025 – March 2026

- Made retrieval follow entity relationships instead of text similarity alone by modeling system structure as a Neo4j knowledge graph linked to its Elasticsearch vector chunks — my first production GraphRAG work.
- Owned the data layer end to end (document creation, chunking, AI-generated chunk summaries, embeddings, dual-store ingestion), made corpus rebuilds repeatable through scripted loaders, and delivered the graph and vector interfaces consumed by the agent layer a teammate owned.

PUBLICATIONS

Vietnamese–English Medical Domain Machine Translation with LLMs and GRPO Optimization Using Verified Rewards

Dang Sy Duy, Nguyen Duy Chien, **Chu Ngoc Vuong**. *Proceedings of the 11th International Workshop on Vietnamese Language and Speech Processing (VLSP 2025)*, Association for Computational Linguistics, October 2025, pp. 377–387. aclanthology.org/2025.vlsp-1.45

- Achieved the strongest overall shared-task performance with Qwen2.5-3B-Instruct — supervised fine-tuning, GRPO, back-translation and a length penalty — beating larger general-purpose systems on terminology fidelity, under a strict 3B parameter cap and limited compute.

RESEARCH

AI for Health: Dementia & Alzheimer's Disease Independent Research

Genome-wide association analysis over ADNI cohorts 2026 – Present

- Building a reproducible genome-wide association pipeline over ADNI genotype and clinical data — PLINK2 quality control, population-structure PCA, logistic and linear association, and inverse-variance-weighted meta-analysis across cohorts.
- Caught two silent data-integrity faults before trusting any result: ADNI1 is on genome build hg18 while ADNI2/ADNI3 are hg19, invalidating naive cross-cohort position matching, and ~46% of ADNI3 variant IDs are chip-manifest names rather than rsIDs — both corrupt downstream association analysis without erroring.

University of Engineering and Technology (UET) Hanoi, Vietnam

Research Student, Human-Machine Interface Laboratory September 2024 – June 2025

- Built an asynchronous real-time pipeline for action classification in indoor surveillance footage, reading directly from live RTSP camera streams.
- Improved accuracy under limited labeled data using semi-supervised cross-model pseudo-labeling over vectorized video-segment features.

HONORS & AWARDS

- **1st Place**, ImageCLEFmed MEDIQA-CORE 2026 — Brain Tumor Subtype Classification
- **1st Place**, Robotics Challenge 2023 — Chiba Institute of Technology, Japan
- **Vietcombank Scholarship** for outstanding academic achievement
- **University Scholarship** for academic excellence — semesters II, VI and VIII

SKILLS

- **Languages** — Python, Java, C++, SQL, Bash
- **LLM / Agentic** — RAG, GraphRAG, hybrid search (BM25 + kNN, RRF), LLM reranking, multi-agent pipelines, prompt engineering, function calling, evaluation harnesses, OpenRouter, OpenAI API, LangChain
- **LLM Adaptation** — QLoRA, supervised fine-tuning, continued pretraining, GRPO / preference-based alignment, back-translation and synthetic data augmentation, Qwen model family
- **ML / CV** — PyTorch, scikit-learn, OpenCV, pandas, NumPy, semi-supervised learning, 3D CNNs
- **Search & Data** — OpenSearch, Elasticsearch, Qdrant, Neo4j, PostgreSQL, MySQL, SQL Server, MongoDB, SQLAlchemy, Alembic
- **Backend & DevOps** — FastAPI, REST API design, Celery, RabbitMQ, async Python, MinIO, Docker, GitLab CI, AWS, Linux, Kubernetes/Helm, pytest, mypy, ruff
- **Languages spoken** — Vietnamese (native), English (professional working proficiency; IELTS 6.5, and taught English to colleagues for 6 months at NTQ Solution)